No.	Status	Date
QSCP-IP-ATU100 Rev 4	Issued	2025-03-20

Piping Inspection Plan - ATU-100 (RBI)

Scope and Responsibilities

This document defines the inspection strategy for the ATU-100 piping systems at the Hamilton Plant, superseding QSCP-IP-ATU100 Rev 3. The scope includes all lines rationalized under the Risk-Based Inspection program, specifically addressing rich amine and reflux services. Jennifer Walsh, as the Inspection Coordinator and API 570 inspector, is responsible for the technical oversight of these activities and ensuring all NDE data aligns with the corporate integrity framework.

All inspection activities must comply with QSCP-CIS-001 Rev 1, which mandates the use of digital data management and specific CML-id rules. The Hamilton Inspection Department is tasked with executing the re-measurement schedules and managing the transition of data into the asset integrity management system. Any deviations from the established intervals or findings below retirement thickness must be documented through the formal nonconformance process.

Rich Amine System Integrity

Line 4-RA-101 is subject to specific monitoring requirements following the discovery of local metal loss at CML-06. This degradation was attributed to flashing downstream of the let-down valve following a 2019 trim change performed without an MOC. Per CIE-FFS-2025-004-R0, a Level 2 Fitness-for-Service assessment has been implemented. The line requires re-measurement in August 2025, with a planned replacement during the August 2026 turnaround.

For the rich amine header, specifically at 6-RA-103-11, CIE-FFS-2025-005-R0 provides the basis for continued operation assuming the let-down valve trim is restored to its original specification. This location must be re-measured in February 2026 to validate the corrosion rates. Additionally, PAUT shall be performed at rich amine fittings to identify potential localized thinning or environmental cracking as per the updated RBI requirements.

Reflux System Nonconformance Management

Inspection of 2-RF-102 identified four CMLs (-04 to -07) that have fallen below the minimum allowable thickness of 2.16 mm. These findings were formally recorded in QSCP-NCR-2024-032, noting that the thinning is localized downstream of the reflux flow-control valve. While 3-RF-101 remains within acceptable limits, the affected sections of 2-RF-102 show a metal loss of approximately 0.9 mm since 2019.

The mitigation strategy for 2-RF-102 requires a re-measurement at a 12-month interval, scheduled for November 2025. The line is slated for replacement during the next scheduled turnaround. Ultrasonic thickness measurements on this A312 TP316 material must be conducted using the specific stainless-steel velocity settings to ensure data accuracy in accordance with API 570 and corporate standards.

Inspection Intervals and NDE Methodology

The inspection intervals for ATU-100 are governed by the RBI criteria established in QSCP-CIS-001 Rev 1. Rich amine services are classified for 3-year intervals, while other Class 2 services such as treated gas are set at 6 years. Remaining life calculations utilize the larger of the long-term and short-term corrosion rates, with the next inspection due at half the remaining life or the class maximum, whichever is more frequent.

NDE personnel must hold certifications to CAN/CGSB-48.9712 or EN ISO 9712. For insulated lines within the CUI temperature range, the program requires profile RT or the installation of insulation windows every 5 years. All thickness data and corrosion mapping results, particularly those involving PAUT at fittings, must be integrated into the Hamilton Plant digital database for ongoing trend analysis.

Sign-off	Name	Function	Date	Signature
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2025-03-20	_____